

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

○



[Lab.1] Cohesive Energy	 <u>3</u>
[Lab.2] Li diffusion activation barrier in bulk structure	 <u>12</u>
[Lab.3] Adsorption energies of Li atoms for different coverage	 <u>18</u>
[Lab.4] Li diffusion activation barrier on the surface	 <u>42</u>
[Lab.5] Interaction of Li Ion with Ethylene Carbonates	 <u>47</u>

[Lab.4] Optimize interfacial structure

Learning Objectives:

→ Confirm the interfacial structure of Li bulk and LiPON bulk through Ab-initio Molecular Dynamics (AIMD) calculations.

Materials: Li/LiPON

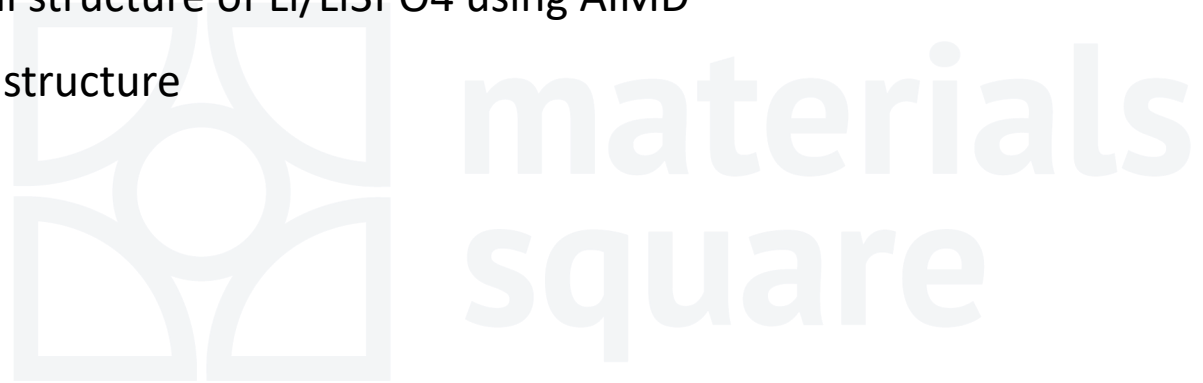


MatSQ Credit: ~ 200 CPU hour (\$ 50)

[Lab.4] Optimize interfacial structure

Li/MoS₂ layer

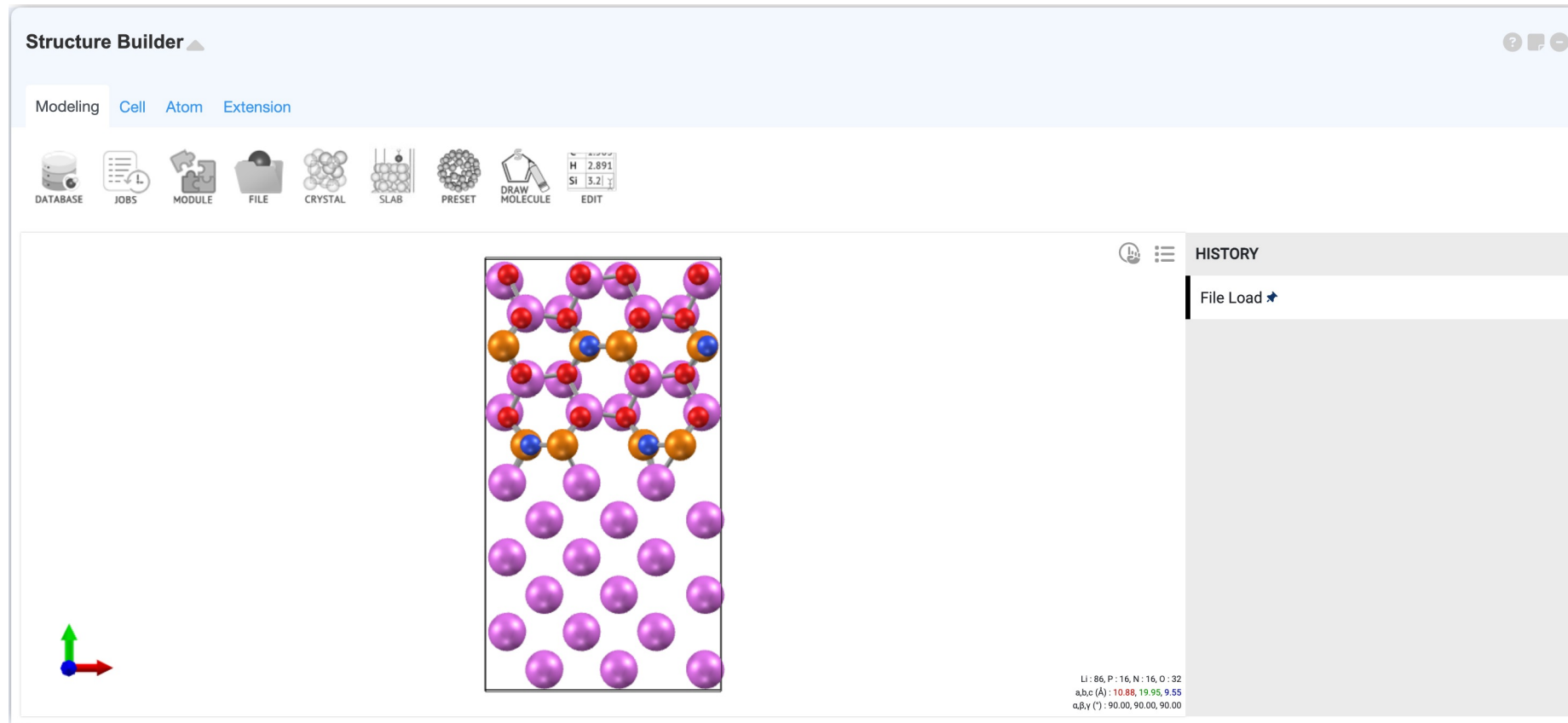
1. Build Li/Li₃PO₄ structure
2. Optimize interfacial structure of Li/Li₃PO₄ using AIMD
3. Analyze interfacial structure



MatSQ Credit: ~ 200 CPU hour (\$ 50)

[Lab.4] Optimize interfacial structure

1. Build Li/Li₃PO₄ structure



[Lab.4] Optimize interfacial structure

2. Optimize interfacial structure using AIMD

Quantum Espresso

PWscf + DOS

Scripting Option : General Atom info Update Structure

&CONTROL

Calculation typemd

MD steps200

Information amountlow

Force threshold (Ry/bohr)0.00038

Time step (a.u.)40

Electric fieldFalse

&SYSTEM

E_{cut}(wfc)(Ry)30

E_{cut}(rho)(Ry)180

Occupationssmearing

DFT FunctionalNone

Gaussian broadening (Ry)0.07

Number of electron spin2 (up,down)

Starting magnetization

Li1

P1

N1

O1

Noncollinear calculation.FALSE.

Van der Waals correctionNone

DFT+U Calculation.FALSE.

noinv.FALSE.

nosym.TRUE.

&IONS

Time integrationverlet

&ELECTRONS

Max iteration step500

SCF must convergeTrue

Mixing modeplain

Mixing beta0.1

Convergence threshold (Ry)0.000001

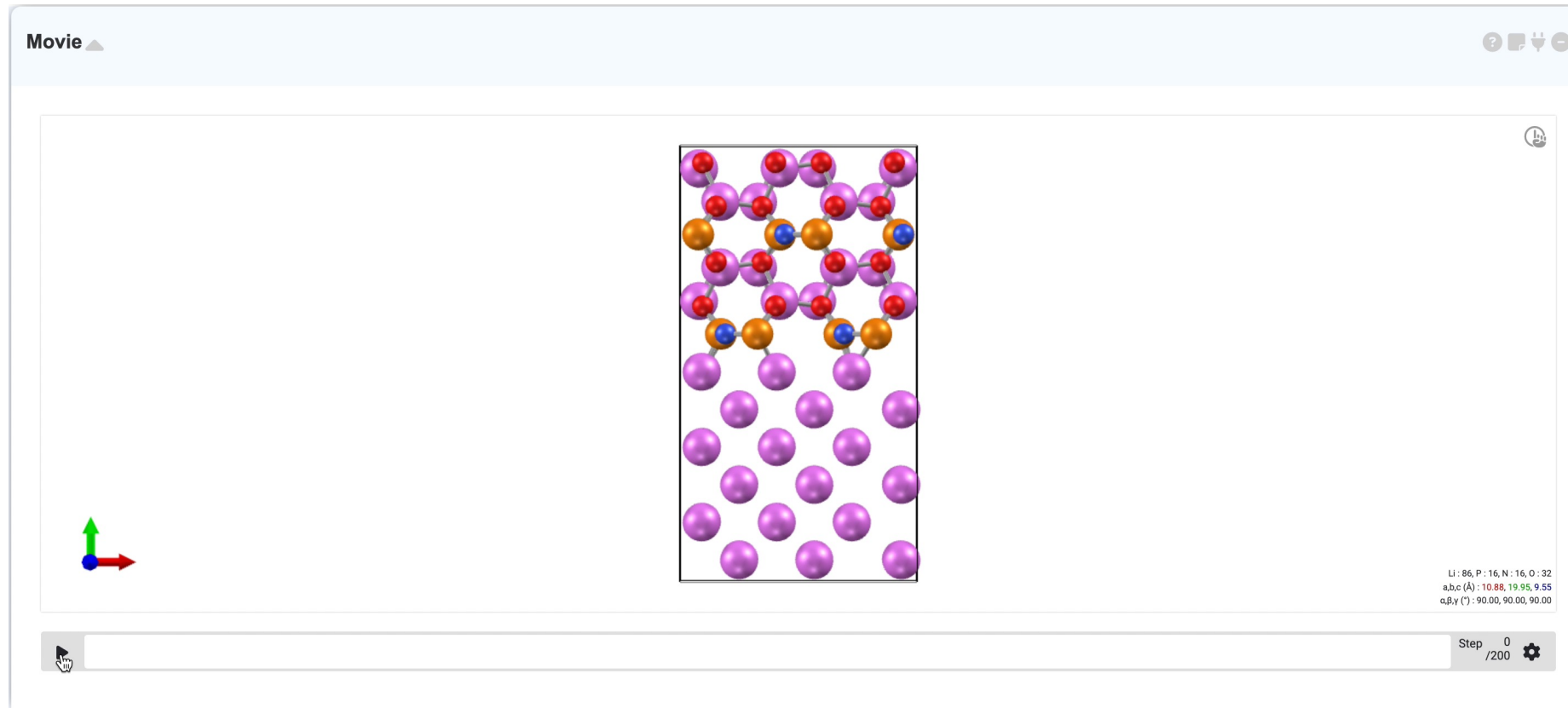
Starting wavefunctionrandom

KPOINTS

SamolniaMonkhorst-Pack

[Lab.4] Optimize interfacial structure

3. Analyze interfacial structure



○

Thank you

○

